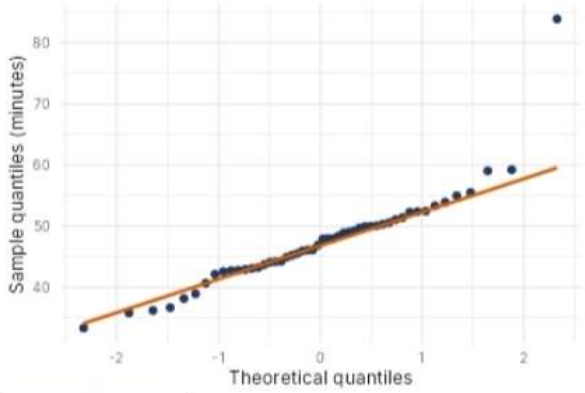
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot C

R.No: 1 Reg. No: 192524157 Name: A Lokesh Kumar

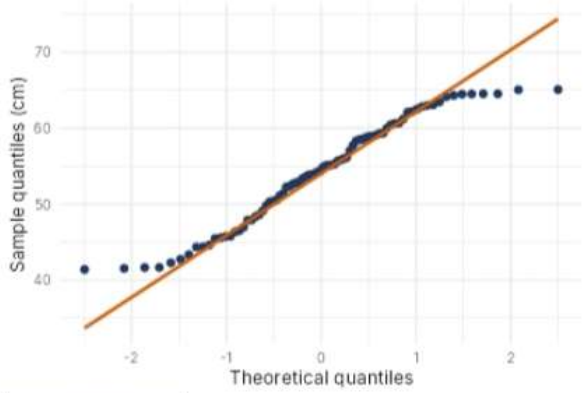
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: commute time (minutes), n = 50



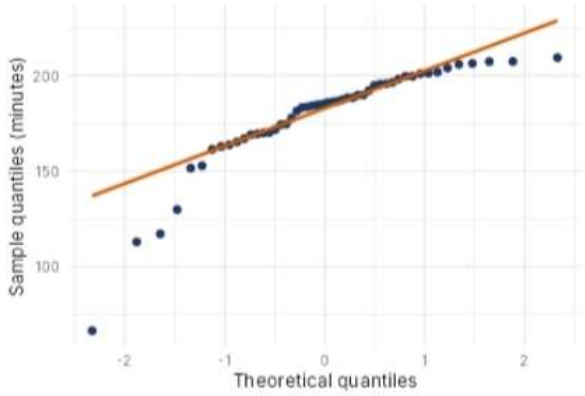
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: plant height (cm), n = 80



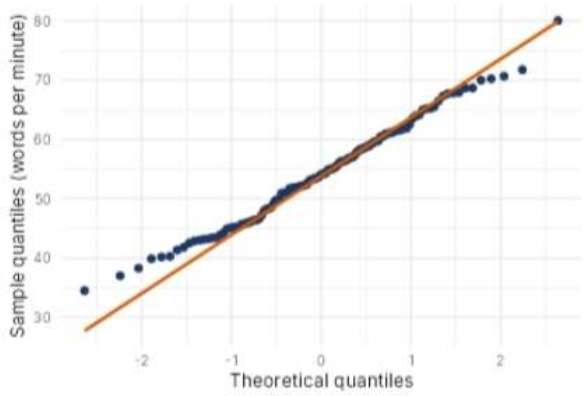
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: daily study time (minutes), n = 50



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: typing speed (words per minute), n = 120



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 2 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Activity - 5

A. Lokesh Kumar
1225241187

Part A: Normality Detection Using Q-Q Plots

Chart 1: Commute Time (minutes)

- Normal? No
- Shape: Right-skewed.
- Evidence: Most points follow the reference line, but the upper tail has an extreme outlier around 84 minutes.

Chart 2: Plant Height (cm)

- Normal? No
- Shape: Light-Tailed.
- Evidence: The points form an S-shaped pattern. Both tails deactivate from the reference line, indicating lighter tails than a normal distribution

Chart 3: Daily Study Time (min)

- Normal? No
- Shape: Left-Skewed
- Evidence: Several points in the lower tail fall far below the reference line, showing a long lower tail.

Chart 4: Typing Speed (words per min)

- Normal? No
- Shape: Light-Tailed.
- Evidence: The points show a slight S-shaped pattern, with deviations at both ends from reference line.

Part B: Answers

B1. chart2

Departure: The data are Right-Tailed and deviate from Normally.

Remedy: Try a suitable transformation, such as a log transformation. If appropriate, or use robust statistical methods instead of mean-and-SD methods

B2. Fill in Blanks

ggplot(df, aes(sample = Value)) + stat_qq() + stat_qq_line()